A Compact and Lightweight Rigid-Tendon Combined Exoskeleton for Hand Rehabilitation

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https://github.com/MengyangGao/hand_exoskeleton

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Abstract

Hand and partial upper extremity impairments are common among stroke and traumatic brain injury survivors, traditionally addressed through manual rehabilitation techniques with varying efficacy. To enhance rehabilitation efficiency, reduce rehabilitator workload, and provide consistent therapeutic outcomes, we developed a compact and lightweight rigid-tendon combined exoskeleton for hand rehabilitation. This study focuses on the design, manufacturing, and testing of the exoskeleton, which consists of two main modules: a rigid finger module and a tendon-driven thumb module. The finger module utilizes a spatial four-bar linkage mechanism, while the thumb module employs a tendon actuation system to achieve multiple degrees of freedom. Advanced bus servo technology was selected for its high precision and closed-loop control capabilities. Additive manufacturing with PLA material was employed to produce the exoskeleton components, ensuring a balance of strength, precision, and cost-effectiveness. We integrated a user-friendly UI interface for control, enhancing the device's accessibility and functionality. The exoskeleton was assembled and subjected to extensive testing. Results demonstrated the device's ability to safely mimic natural hand movements, generate a maximum gripping force of 23.64 N—exceeding the design requirement by 18.2%—and offer customizable movement speeds from 0.2 to 2 seconds. Each finger's movement could be controlled independently or in combination with one degree of freedom (DOF), and the thumb module achieved two DOF. The exoskeleton's lightweight and integrated design enhance user comfort and portability, making it a viable solution for both medical rehabilitation and assistive applications.

Keywords: Hand Exoskeleton, Rehabilitation Robotics,Additive Manufacturing, Bus Servo, Four-Bar Linkage, Tendon-Driven Thumb.

1 Introduction

1.1 Project Background and Target

Hand and partial upper extremity impairments are common among stroke and traumatic brain injury survivors. At present, the traditional rehabilitation treatment mainly relies on medical rehabilitators, and the therapeutic effect is affected by the experience and energy of the rehabilitators, so the therapeutic efficiency is relatively low. In order to improve the efficiency of rehabilitation treatment, make up for the shortages of rehabilitators, and reduce the work intensity of rehabilitators, corresponding rehabilitation treatment equipment has emerged . Exoskeleton belongs to wearable equipment, which can be used for limb rehabilitation. The use of exoskeleton for rehabilitation training has gradually become the mainstream of rehabilitation training [1]. The research of exoskeleton robots is developing in the direction of lightweight and intelligence. The hand exoskeleton can be fixed on the patient's hand to drive the fingers to do flexion and extension movements and other rehabilitation training[2]. The lightness, comfort and safety of the exoskeleton are particularly important in the rehabilitation training process. Finger exoskeleton is the basic module of hand exoskeleton and the main module to realize the function, so the design of finger exoskeleton directly affects the realization of hand exoskeleton function.

The project aims to design an assistive hand exoskeleton glove that enhances strength and posture control in the hand, facilitating the completion of complex manual tasks. This wearable robotic system, encompassing the wearer's wrist, palm, and fingers, is designed for rehabilitation and assistance purposes. It has significant applications in the medical field, aiding in the recovery of hand functionality and improving quality of life, and in the industrial and construction sectors, where it can reduce workload and minimize the risk of occupational injuries. The design incorporates various transmission mechanisms, such as linkages to achieve the required motion and force, with a focus on safety, ensuring that the device prevents excessive motion and counterproductive joint movements.

1.2 Literature Review

The typical human finger structure consists of three joints: the distal interphalangeal joint (DIP), the proximal interphalangeal joint (PIP), and the metacarpal interphalangeal joint (MCP). The entire finger consists of multiple bones, muscles, blood vessels and nerves, which provide a high degree of dexterity and fine manipulation. The design of the hand exoskeleton needs to fully consider the physiological characteristics and movement patterns of the fingers to ensure that the device can coordinate with the human hand to achieve the best enhancement and assistive effect. From the perspective of flexibility, mainstream hand exoskeletons can be categorized into rigid hand exoskeletons and flexible hand exoskeletons, among which the two mainstream solutions for flexible hand exoskeletons are gas-driven hand exoskeletons and lasso-driven hand exoskeletons. In the literature research section, we classify hand exoskeletons into four categories, including: Rigid hand exoskeletons, Tendon hand exoskeletons, Pneumatic hand exoskeletons, and Rigid-Soft hybrid hand exoskeletons.

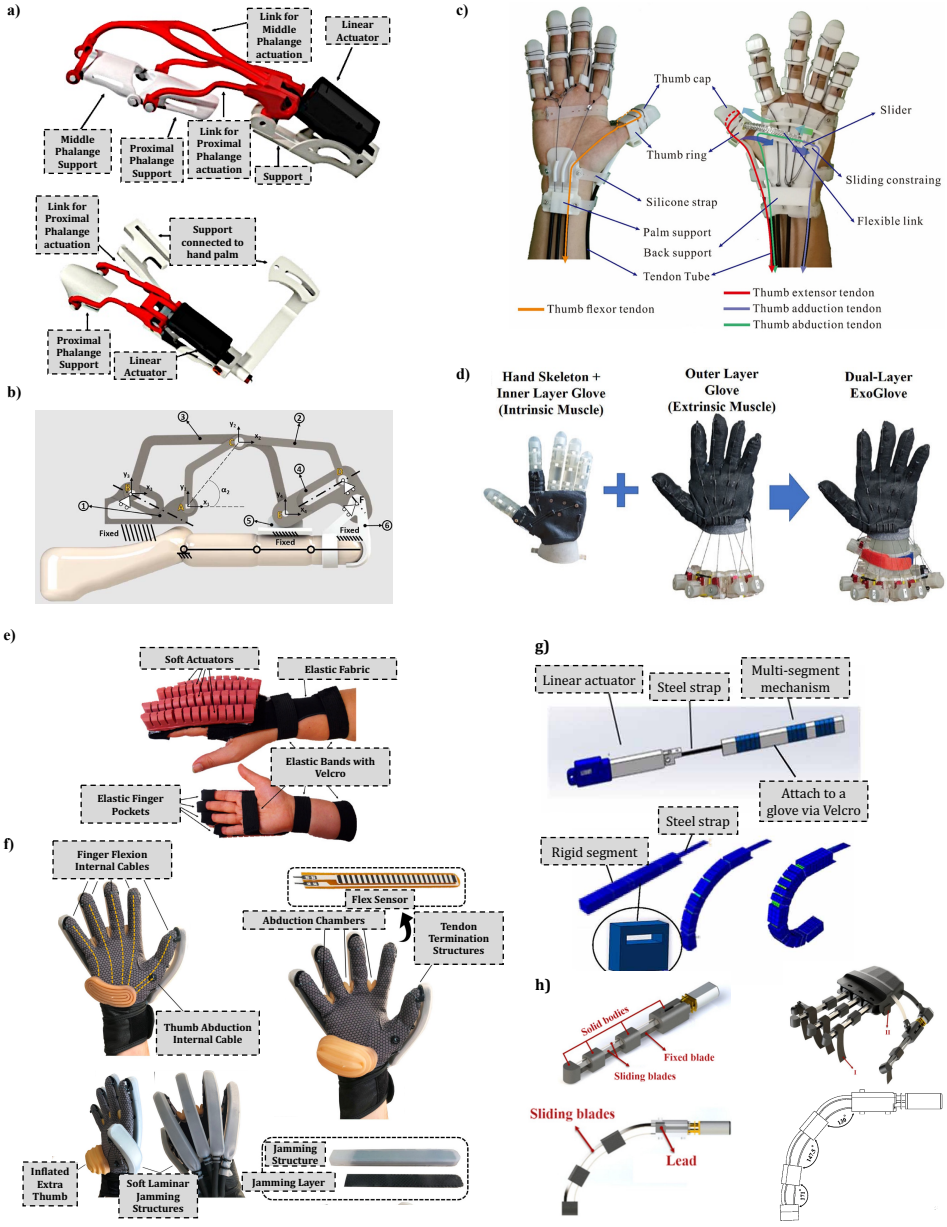

Figure 1: Different types of hand exoskeletons.

The various types of hand exoskeletons each bring unique advantages; rigid exoskeletons ensure precise control and strong grasp forces, tendon exoskeletons offer portability and biomimetic movement, pneumatic soft exoskeletons use lightweight materials for safe power assistance, and combined

rigid-soft exoskeletons balance natural movement with control precision. However, these systems also face significant challenges. Rigid exoskeletons can be bulky and may cause joint pain due to misalignment. Tendon systems risk discomfort from uneven tendon tension and may compromise the natural feel of hand movements. Pneumatic soft exoskeletons, while lightweight, suffer from control complexities due to the non-linearity of soft actuators and require tethering to bulky equipment, limiting their practicality. Finally, combined rigid-soft exoskeletons confront design and integration difficulties, particularly in merging different material properties and managing interactions between rigid and flexible components.

More detailed literature information is covered in the first report of this project (Literature Review, File S1, Supporting Information).

2 Design Proposal

In previous report, we proposed that 3D printing technology is used to produce the hand exoskeleton. The entire exoskeleton of the hand is composed of three main mechanisms: a **thumb module mechanisms**, **finger module mechanisms** and a **wrist model mechanisms**.

2.1 System Overview

This block diagram shown in figure 1 presents an integrated system comprising two core modules: the Rehabilitation Module and the Instructional Module, each outfitted with motorized mechanisms facilitating finger movement. Both modules are constituted by actuators designated for the 'Thumb' and 'Other fingers', which are interconnected through 'Linkage' and 'String' components. This design ensures the precise manipulation of the digits.

Central to the system's operation is the ESP32-NOW protocol, enabling wireless communication between the modules. The ESP32-NOW serves as a pivotal element, leveraging the Wi-Fi capability intrinsic to the ESP32 chipset, thus negating the necessity for a conventional Wi-Fi network infrastructure.

The Rehabilitation Module is primarily designed to execute therapeutic movements, directly driving the actuators through a complex sequence of commands. Concurrently, the Instructional Module mirrors the mechanical structure of its counterpart but is distinct in its purpose, functioning as a control interface for the operator, presumably a clinician or therapist, to demonstrate desired movements.

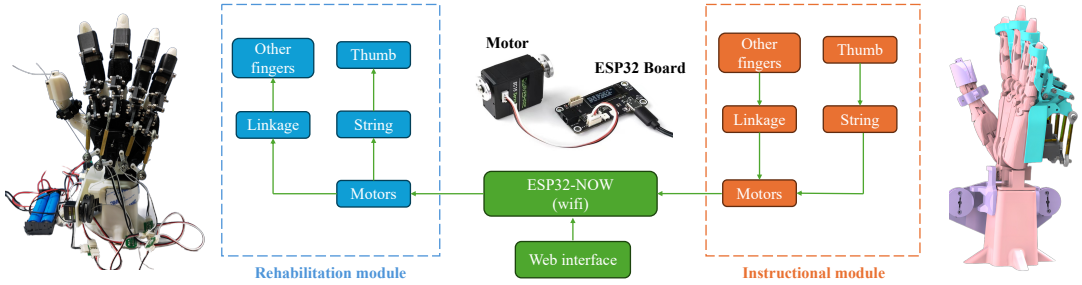

Figure 2: Overview of the proposed system

2.2 Mechanical Design

In previous Mid-Report (Design Proposal, File S2, Supporting Information), we proposed that 3D printing technology is used to produce the hand exoskeleton. The entire exoskeleton of the hand is composed of three main mechanisms: a **thumb module**, **finger module** and a **wrist module**. Herein,

we modified the classification and sorted the hand exoskeleton into two sections: **Rigid finger module** and **Tendon thumb module**.

2.2.1 Rigid Finger Module

1. Modelling Principles

In the finger module, the PIP and MCP joints are located between the three phalanges of the remaining four fingers. To ensure alignment between the center of the MCP joint and the rotation center of the mechanism, and to prevent interference with the human hand and secondary injuries to the soft tissue, a parallelogram remote projection center mechanism is utilized to actuate the MCP joints of these four fingers. The driving mechanism consists of a spatial four-bar mechanism, while the executing mechanism comprises a planar multi-bar mechanism. Kinematic analyses of both mechanisms are conducted independently. The spatial RSSR mechanism is constructed with two ball hinges, two rotating pairs, and four bar members, with their relative positions depicted in Figure 3a and 3b [3].

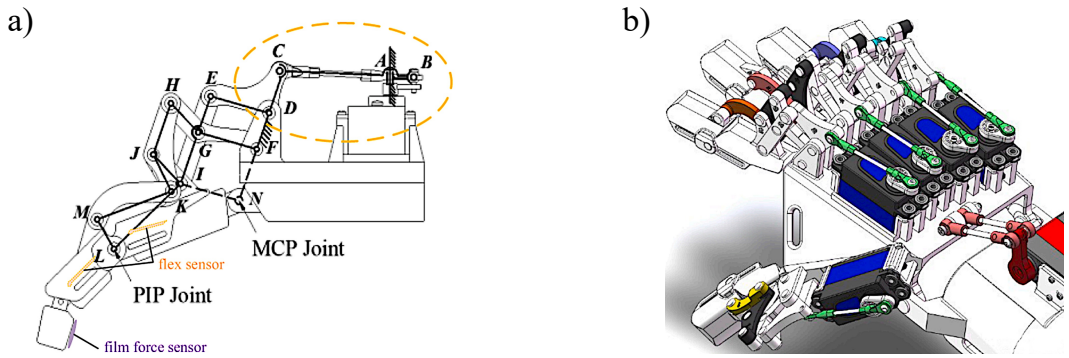

Figure 3: Referenced rigid linkage structure [3].

The driving mechanism consists of a spatial four-bar linkage, while the executing mechanism comprises a planar multi-bar linkage. Kinematic analyses for these mechanisms are performed separately. The spatial mechanism includes two spherical joints, two revolute pairs, and four connecting bars. Their relative positions are depicted in Figure 4. In this figure, the coordinate systems for the driving link AB, driven link CD, and rack AD are denoted as $o_1x_1y_1z_1$, $o_3x_3y_3z_3$, and $o_4x_4y_4z_4$ respectively. The intersection angles for the driving link AB and driven link CD are θ_1 and θ_3, respectively. The parameters s_3, h, and s_4 represent the lengths of segments Dd, ad, and Aa as shown in Figure 2, respectively. The lengths of links AB, BC, and CD are denoted as l_1, l_2, and l_3 respectively. The angle between z_1 and z_4 is α_4. The coordinates of point B are $[l_1\,0\,0]$ in the coordinate system $o_1x_1y_1z_1$, and its coordinates in the system $o_4x_4y_4z_4$ are as follows:

$$\begin{bmatrix} x_4^{(B)} \\ y_4^{(B)} \\ z_4^{(B)} \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} l_1\cos\theta_1 \\ l_1\sin\theta_1 \\ s_4 \end{bmatrix} = \begin{bmatrix} l_1 \\ 0 \\ s_4 \end{bmatrix} \tag{1}$$

Similarly, the value of point C in the coordinate system $o_4x_4y_4z_4$ is as follows:

$$\begin{bmatrix} x_4^{(C)} \\ y_4^{(C)} \\ z_4^{(C)} \end{bmatrix} = \begin{bmatrix} l_3\cos\alpha_4\sin\theta_3 + s_3\sin\alpha_4 \\ -l_3\sin\alpha_4\cos\theta_3 + s_3\cos\alpha_4 \\ -h + l_3\cos\theta_3 \end{bmatrix} \tag{2}$$

As the length of l_2 is constant in the space mechanism, its constraint equation is given below:

$$\left(x_4^{(B)} - x_4^{(C)}\right)^2 + \left(y_4^{(B)} - y_4^{(C)}\right)^2 + \left(z_4^{(B)} - z_4^{(C)}\right)^2 = l_2^2 \tag{3}$$

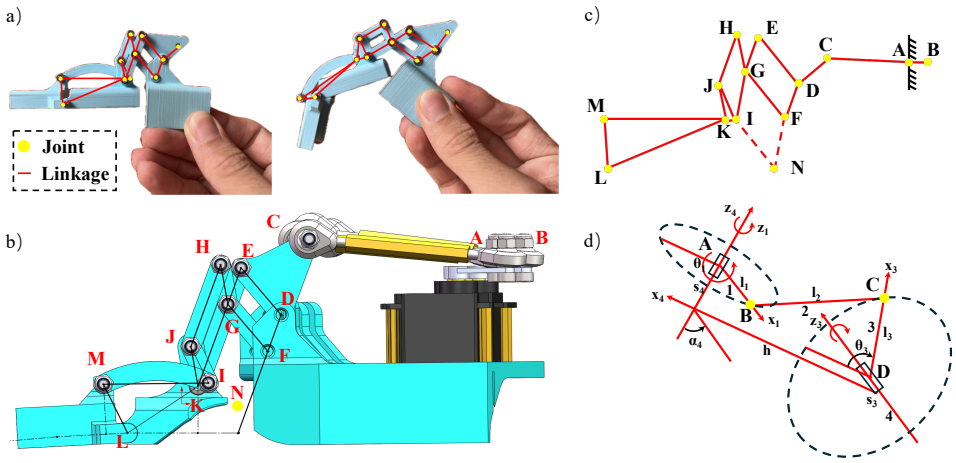

Figure 4: Schematic of the finger structure and the kinematic diagram of the finger part of exoskeleton.

According to Eq.(1)-Eq.(3), the equation of θ_3 can be obtained as below:

$$\rho_1 \cos\theta_3 + \rho_2 \sin\theta_3 + \rho_3 = 0 \tag{4}$$

where

$$\rho_1 = -\cos\alpha_4 \sin\theta_1 + \frac{s_4 \sin\alpha_4}{l_1},$$

$$\rho_2 = h - \cos\theta_1,$$

$$\rho_3 = -s_3 \sin\alpha_4 \sin\theta_1 + h\cos\theta_1 + \frac{1}{2l_3}\left(-l_2^2 + l_1^2 + l_3^2 + h^2 + s_3^2 - 2s_3 s_4 \cos\alpha_4\right).$$

Where ρ_1, ρ_2, and ρ_3 are used to simplify the equation of θ_3. The detailed expression of ρ_1, ρ_2, and ρ_3 are shown in Eq.(4). Then, the solution of θ_3 is as follows:

$$\theta_3 = 2\arctan\left(\frac{\rho_1 \pm \sqrt{\rho_1^2 + \rho_2^2 - \rho_3^2}}{\rho_2 - \rho_3}\right) \tag{5}$$

According to Eq.(5), it is known that the solution of θ_3 has two values. By analyzing the position shown in Figure 5 and the continuity of the motion of the four-bar linkage, it can be determined that a positive value should be taken at this time. The angle of the swing bar CD represented by θ_3 is a function of the angle of the swivel arm AB represented by θ_1, and it has also been verified that the rotational degree of BC does not affect the movement of the swing bar CD.

For the MCP joint, given the space constraints on either side of joint node N, two parallel double crank mechanisms, EDGF and JHGI, were employed to externally actuate the joint. With parallelogram EG parallel to DF and segment EGI equal to segment DFN, EKND forms a parallelogram, ensuring lengths IN, ED, and GF remain constant. Thus, angle $\angle$JIN is maintained constant, allowing the first phalanx to pivot around the MCP joint. For the PIP joint's second phalanx control, node ML is part of the LJKM four-bar linkage, with LJ serving as the frame and ML moving with HJK.

According to the geometric properties of the double parallelogram in the four-sided linkage, the angles of all bars parallel to the frame DF remain unchanged in this configuration. Therefore, the angle θ_N can be expressed as a function of θ_3:

$$\theta_N = \theta_3 + \alpha_5 + \alpha_{93} \Rightarrow \theta_N = f(\theta_3) \tag{6}$$

Since α_5 and α_{93} are constants, θ_N is a function of θ_3, as depicted in Fig.3. Furthermore, the following relationships hold:

$$\begin{aligned} \theta + \alpha_7 + \alpha &= \pi \\ \theta_3 + \alpha_5 - \pi &= \alpha \\ 2\pi - \alpha_{92} + \theta &= \theta_9 \end{aligned} \tag{7}$$

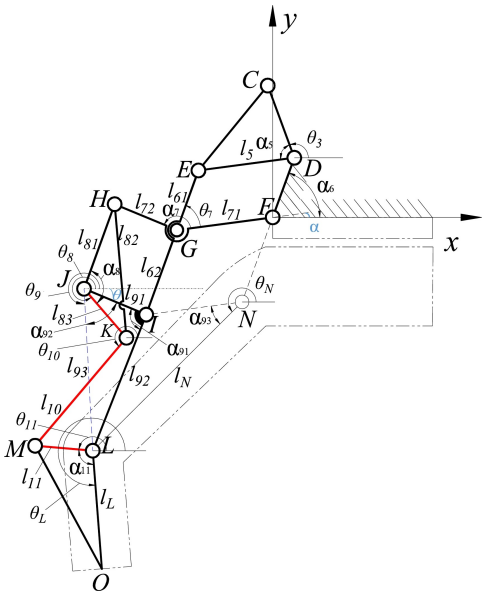

Figure 5: The setting of coordinate axis of the finger part [3].

Consequently, the solutions for θ_9 and θ_8 are functions of θ_3 with constants α_6, α_7, α_8, and α_{92}:

$$\theta_9 = \theta_3 + \alpha_5 + \alpha_7 - \alpha_{92} \tag{8}$$

$$\theta_8 = 2\pi - \alpha_8 + \alpha_6 \tag{9}$$

In the four-bar mechanism JILMKJ, the vector sum of link lengths in the closed loop must equal zero, verifying the loop closure condition:

$$J\vec{K} + K\vec{M} = J\vec{L} + L\vec{M} \tag{10}$$

The vector equation can be simplified into two scalar equations as follows:

$$l_{83}\sin\theta_8 + l_{10}\sin\theta_{10} = l_{93}\sin\theta_9 + l_{11}\sin\theta_{11} \tag{10}$$

$$l_{83}\cos\theta_8 + l_{10}\cos\theta_{10} = l_{93}\cos\theta_9 + l_{11}\cos\theta_{11} \tag{11}$$

Eq.(11) can be reformulated as:

$$A\cos\theta_{11} + B\sin\theta_{11} + C = 0 \tag{12}$$

where:

$$A = l_{93}\cos\theta_9 - l_{83}\cos\theta_8$$

$$B = l_{93}\sin\theta_9 - l_{83}\sin\theta_8$$

$$C = \frac{A^2 + B^2 + l_{11}^2 - l_{10}^2}{2l_{11}}$$

Solving this equation provides the solution for θ_{11}, which is a function of θ_8 and θ_9, and is represented as $F(\theta_8, \theta_9)$:

$$\theta_{11} = 2\arctan\left(\frac{B - \sqrt{A^2 + B^2 - C^2}}{A - C}\right) \Rightarrow \theta_{11} = F(\theta_8, \theta_9) \tag{13}$$

According to Fig. 3, the angle θ_L is as follows:

$$\theta_L = \theta_{11} + \alpha_{11} \Rightarrow \theta_L = g(\theta_3) \tag{14}$$

The angle θ_L is a function of θ_3, denoted as $g(\theta_3)$. As depicted in Fig. 3, the lengths of the proximal phalanx and the middle phalanx correspond to l_N and l_L. The coordinates for points L and O:

$$x_L = l_N \cos\theta_N + l_L \cos\theta_L \tag{15}$$

$$y_L = l_N \sin\theta_N + l_L \sin\theta_L \tag{16}$$

$$x_O = l_N \cos\theta_N \tag{17}$$

$$y_O = l_N \sin\theta_N \tag{18}$$

Eq.(8), (9), (14), and (15) demonstrate that the finger joint angle is a function of the input angle θ_3, confirming that the motions of both the MIP and PIP joints can be controlled with a single driving link. The kinematic analysis indicates efficient motion transfer from the RSSR space to the linkage mechanism with only one moving part, establishing that the manipulator training mechanism can coordinate two joint motions using a single input.

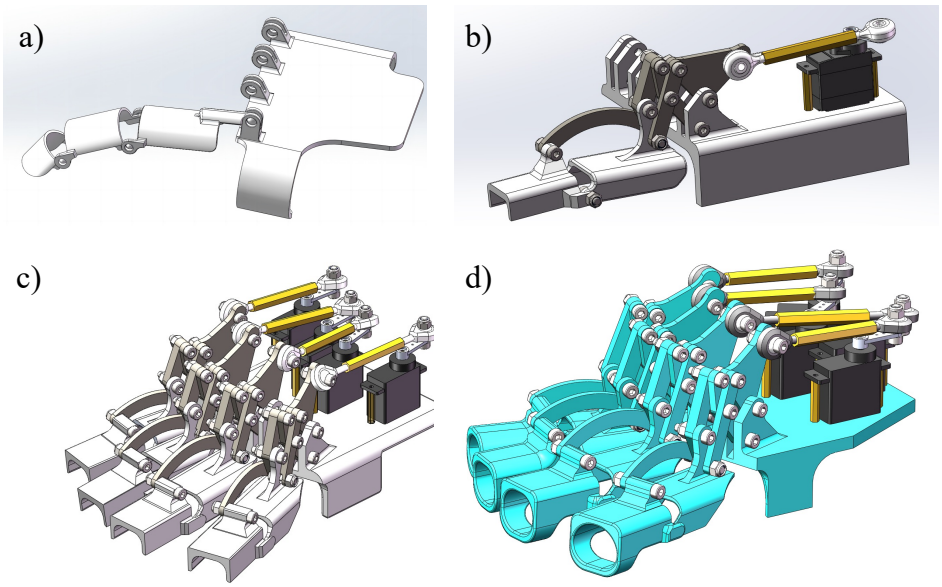

Figure 6: Iterative Design Process of Finger Assistive Exoskeleton

2. Model Iteration

The iterative design process of our finger assistive exoskeleton includes four stages, each improving upon the last to enhance functionality, user comfort, and ease of use, shown in figure 6. We utilized the Solidworks to model the whole finger model, which can transform the circular motion of the servo into linear motion of the finger.

This **first** stage involved creating a basic conceptual model without a transmission mechanism. The focus was on establishing the fundamental structure and understanding the basic requirements for finger assistance without the complexities of power transmission. The **second** stage introduced a linkage system for the non-powered section of the exoskeleton. This addition aimed to provide a more precise movement framework, laying the groundwork for more advanced powered mechanisms in subsequent stages.

In the **third** stage, the design was scaled to match human hand dimensions, incorporating adjustable straps to improve finger fit and reduce interference. Structural reinforcements were added to enhance durability, and transmission dimensions were modified to increase the output torque. These improvements aimed to make the exoskeleton more robust and effective in assisting finger movements.

The **final** stage focused on eliminating inter-finger coupling interference by introducing end rings, which also reduced the reliance on straps. This design enhancement significantly improved wearability

and ease of use, making the exoskeleton more practical for everyday use. The structural adjustments ensured that the exoskeleton could be easily donned and doffed while maintaining a high degree of functionality.

2.2.2 Tendon Thumb Module

The primary joints of the thumb include the IP and the MCP. The thumb's abduction and adduction movements are facilitated by tendon actuation. Specifically, the adduction movement is enabled by a single tendon, illustrated as a yellow line in Figure 6a. Additionally, the thumb can perform flexion movements through the action of the flexor pollicis longus tendon. This functionality is mimicked by the thumb flexor tendon, depicted as a green line in figure 2a, which extends across both the MCP and IP joints. Figure XX shows the 3d thumb model in Solidworks, which can correspond to the thumb finger properly. [4].

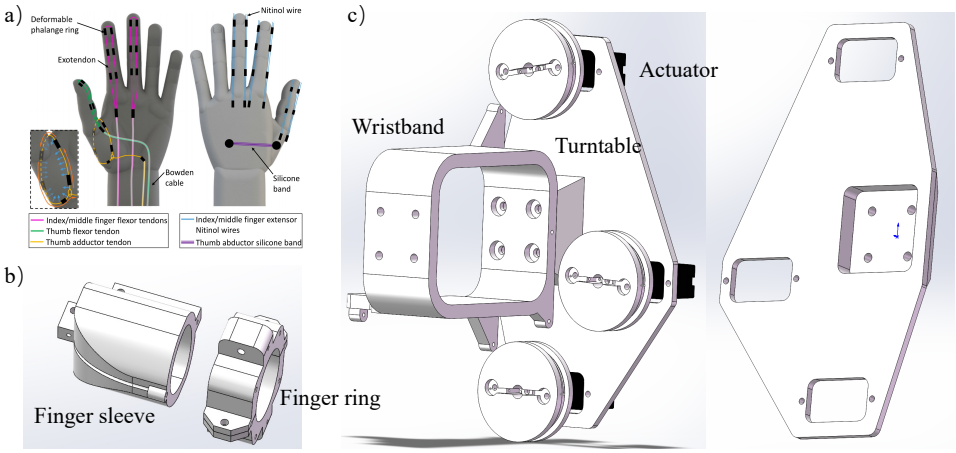

Figure 7: The setting of coordinate axis of the finger part.

In the design of the electronic control system for the thumb, three servo motors work in coordination to achieve precise control with two degrees of freedom. Servo motor 2 is responsible for controlling two wires, while servo motors 1 and 3 each control one wire independently. To ensure consistent initial movements, the angles of the three SC series servo motors are first adjusted to the midpoint angle using an ESP32 controller, serving as the starting position. This setup prepares the system for executing various action plans. During the execution of each action, the servo motors' feedback angle data is tested and recorded one by one to ensure that each servo motor can accurately rotate according to the predetermined action plan upon receiving commands. This precise rotation drives the mechanical structure to simulate the thumb's movement trajectory.

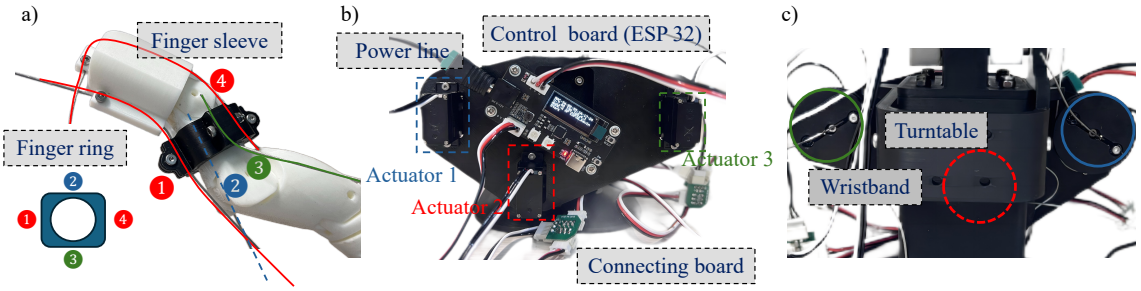

Figure 8: The setting of coordinate axis of the finger part [3].

In the latest iteration of the mechanical assistive exoskeleton, significant improvements have been made to the wrist section. The previous version utilized screw fastenings to secure the wrist mechanism. However, this approach often proved cumbersome and less adaptable to the varied needs of

patients. The new design replaces the screw fastenings with adjustable straps featuring Velcro closures. This modification not only simplifies the process of putting on and taking off the exoskeleton but also offers a more customizable fit for different wrist sizes. Patients can easily adjust the tightness of the straps, ensuring a more comfortable and secure experience during use. This upgrade enhances overall usability and better aligns with the practical needs of patients, making the exoskeleton more user-friendly and efficient in daily rehabilitation tasks.

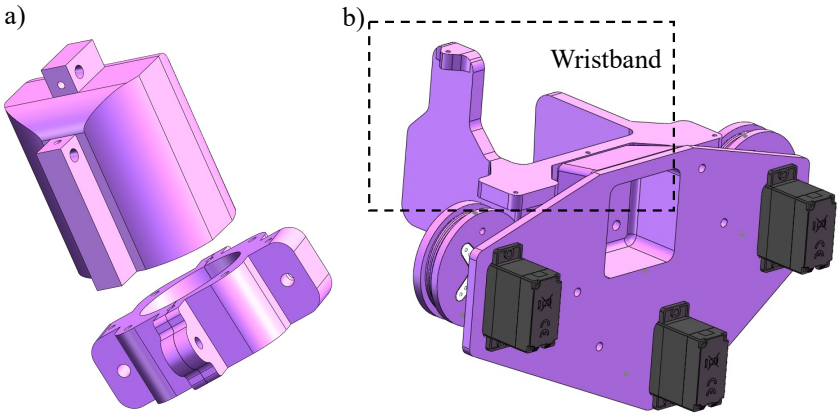

Figure 9: Upgraded wrist mechanism of the exoskeleton.

2.3 Software Control

The rehabilitation device utilizes an ESP32 microcontroller unit (MCU) to communicate precise movement commands to seven servos that control the fingers and thumb. Each SC09 servo is equipped with a Proportional-Integral-Differential (PID) controller, optimizing the control over the servo's rotational angle. This mechanism ensures that the movements are both accurate and smooth, which is crucial for the efficacy of rehabilitation exercises. The ESP32 sends a series of ID-Position-Speed pairs

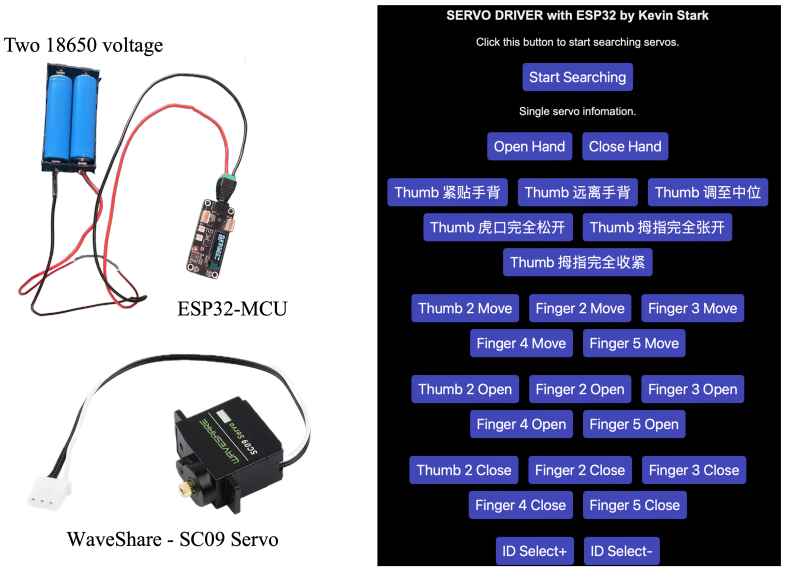

Figure 10: Schematic of the hardware and web interface.

to the servos, each pair dictating the target position and the movement speed for a specific servo identified by its ID. This setup allows for a coordinated control over the servos, enabling the mechanical structure of the exoskeleton to assist in movement or perform specific rehabilitation training sequences. The power requirements for the ESP32 and the servos are met by connecting two 18650 batteries in series, providing a total of 7.4V, which falls within the operational range of 6V to 12V for the ESP32.

The connection is established via a 5.5 x 2.1 mm DC jack on the ESP32, ensuring a steady and reliable power supply for the duration of the rehabilitation activities. The servos are interconnected using a bus topology with their signal lines connected to the 3-pin interface of the ESP32. This setup facilitates synchronized control commands across all servos, which is crucial for complex movement patterns involving multiple fingers. The program for the ESP32 is uploaded through a USB Type-C port, which is also used for debugging and firmware updates.

Advanced control strategies are employed, especially for the thumb, which uses a three-servo, four-wire dual-degree-of-freedom mechanism. The coordination among the servos involves complex control logic to manage the interaction between the servos to mimic natural thumb movements. For instance, simultaneous and opposing actions by two servos are required to achieve the desired motion trajectory without overloading the mechanical structure.

The web interface depicted in the image is designed to control a finger exoskeleton system via an ESP32. This interface is intuitively organized, offering a user-friendly environment for manipulating individual servos associated with the fingers and thumb of the exoskeleton. The main features include buttons for "Open Hand" and "Close Hand," which provide quick access to predefined movement patterns. Additional controls are segmented into specific movements for each digit, such as "Thumb Move," "Finger 2 Move," and others, allowing precise control over the movement, opening, and closing of each finger. This modular approach facilitates detailed command execution which is essential for targeted rehabilitation exercises. The interface also includes functionality for searching servo IDs and adjusting them directly, enhancing the flexibility and adaptability of the system to different user needs.

3 Controlling and Manufacturing Strategy

As previously introduced, the whole device comprises two modules: the Finger module and the Thumb module. The proposed solution for the finger consists of a rod drive system with several 3D printed parts assembled respectively. Opening/closing motion of the hand is actuated by means of four finger modules with one bus servo per module. The thumb module controls multiple degrees of freedom through three bus steering gear pull-lock actuations. Before tackling the experimental evaluation, we describe the detailed manufacturing process of the exoskeleton.

To utilize the knowledge acquired from coursework, we planned to adopt a linkage mechanism to drive the hand-assist exoskeleton. Considering the thumb module's requirement for achieving two degrees of freedom, and given the complexity and potential dead zones using a rigid structure, we employed two different drive schemes for the fingers and thumb parts, respectively, and designed corresponding SolidWorks models. During verification, we conducted multiple iterations to optimize the transmission efficiency of the linkages and minimize the exoskeleton's motion dead zones.

3.1 Control Method

Regarding servo selection, we investigated both Pulse Width Modulation (PWM)servos and bus servos to determine the most suitable control scheme for the hand-assist exoskeleton system. PWM servos are controlled by PWM signals. A microcontroller generates PWM signals and sends them to the servo, controlling its angle by adjusting the signal's pulse width. This control principle is based on the combination of a motor and a feedback control system, with the servo containing a DC motor, a reduction gear, and a closed-loop feedback system. The servo receives PWM signals, generates a bias voltage, and triggers the motor to move the potentiometer. The motor stops when the voltage difference is zero, achieving the servo effect. PWM servos are simple to use, widely applied in model aviation, and can quickly respond to angle changes, making them suitable for scenarios requiring rapid angle adjustments. However, their precision is relatively low, usually only rotating from the initial position to the target position without precise positioning or feedback capabilities. Additionally, the

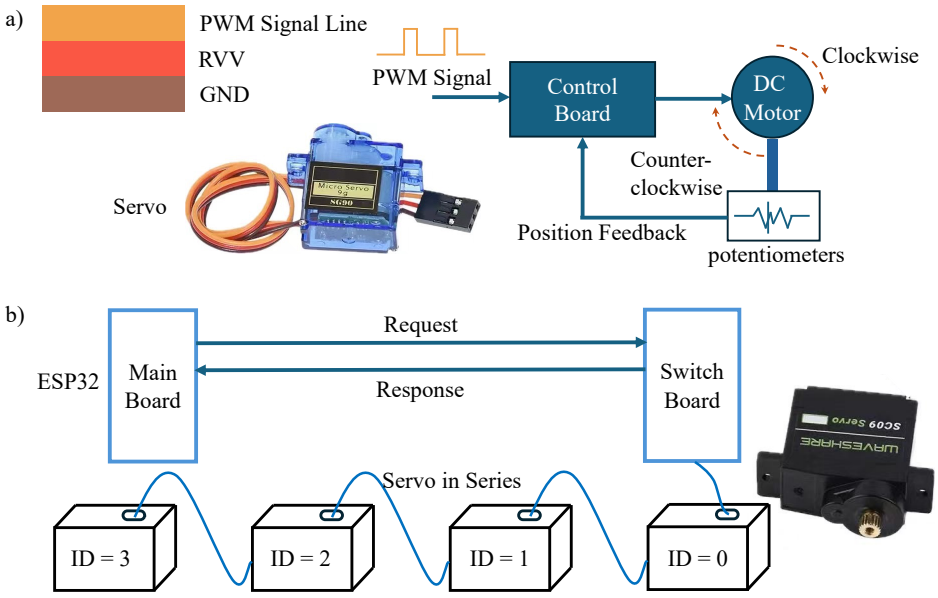

Figure 11: Comparison of PWM serco and bus servo.

complex wiring to the mainboard can lead to issues, and their lifespan is limited due to common gear wear and backlash problems.

In contrast, bus servos use serial communication to control servo motion through specific protocol data transmission, comprising a controller, servo, and bus. The controller provides a control interface, allowing commands to be sent to the servo via buttons, knobs, or computer software. The bus servo works by receiving commands and adjusting parameters such as position, speed, and torque. Its advantages include closed-loop control and feedback functions, enabling real-time monitoring of parameters like voltage, speed, temperature, position, and current. Furthermore, bus servos can control multiple servos in series, each with a unique ID, offering higher precision and multiple controllable parameters, robust performance, and the ability to read the real-time position of specified servos for teaching purposes. They are suitable for applications in robotic arms and smart home systems.

After comprehensive comparison, we chose bus servos for the hand-assist exoskeleton's drive system. The high precision and controllable parameters of bus servos better facilitate the multi-degree-of-freedom movement of the thumb, while their closed-loop control and real-time feedback functions meet our requirements for precise exoskeleton control and teaching functionality.

3.2 Additive Manufacturing

After completing the SolidWorks modeling and part selection, we used 3D printing technology to manufacture the modeled parts. Considering the mechanical performance requirements for strength and hardness, precision and surface quality demands, overall system weight limitations, and an economical and environmentally friendly approach, we selected PLA as the printing material. During model slicing, we set a sparse infill density of 20%, adjusted the support distance and printing speed, and after printing, removed supports and polished the parts. Additionally, we printed a hand model provided by the teaching assistant and scaled it to 90% of human hand size to facilitate testing the exoskeleton's various parameters.

3.3 Assembly and Debugging

We assembled the printed models. In the finger module, we used ball joints to link the servo and linkage, converting the servo's torque into the linkage's driving force, which actuated the four-finger motion. We adjusted the linkage length to maximize the finger bending. In the thumb module, we

3D printed matching servo horns to convert the servo's torque into the cable's driving force, enabling the thumb's two degrees of freedom motion. After assembly, we proceeded to debugging and subsequently tested the results to ensure proper functionality and performance.

4 Results and Tests

4.1 Safety and Travel Range

The entire movement process ensured that the exoskeleton did not overextend or force the user's hand into unnatural positions, thus preventing hyperextension and reverse joint movements. The device successfully achieved full contraction and extension, accurately mimicking and replicating human hand movements, shown in figure 12.

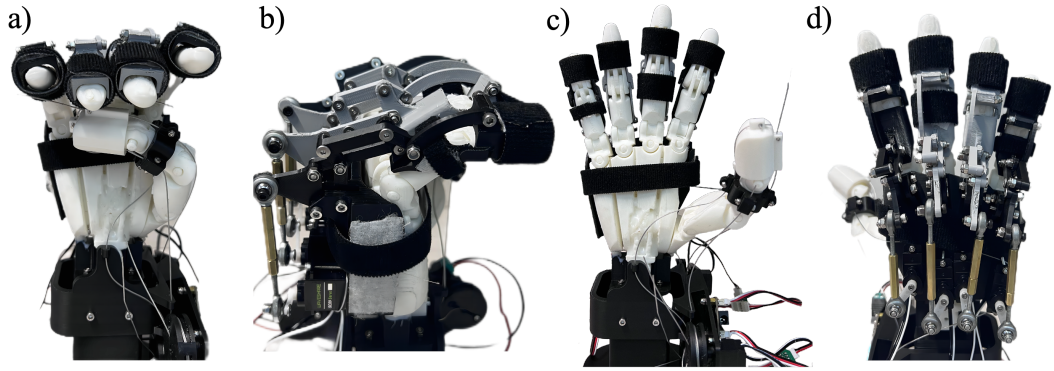

Figure 12: Schematic of two states of exoskeleton, (a) and (b) for closed; (c) and (d) for open.

4.2 Driving Torque

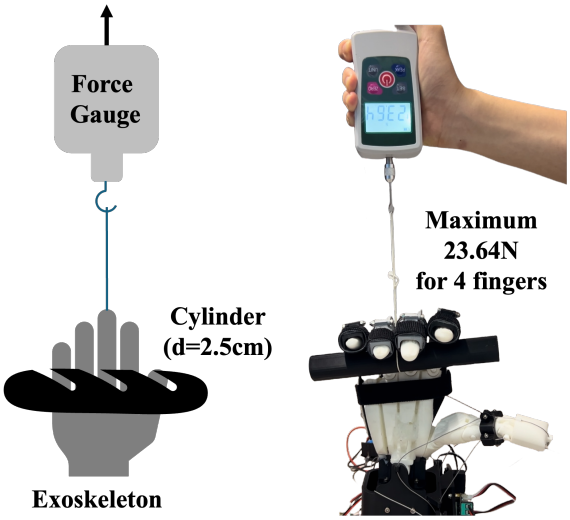

Figure 13: Experiment of the test of the grip force.

Testing showed that our hand-assist exoskeleton could grip a rod with a 2.5 cm outer diameter and suspend a 20 N load while maintaining the fingers in a bent position, meeting the design requirements. Additionally, to gain a more detailed understanding of the exoskeleton's performance, we used a dynamometer to test the maximum gripping force of the exoskeleton. Results indicated that the four-

finger section of our assistive exoskeleton could support a 23.64 N load when gripping a rod with a 2.5 cm outer diameter, exceeding the design requirement by 18.2%, shown in figuer 13.

4.3 Speed

Testing demonstrated that the speed of transitioning from an extended hand to a fully grasped state was controllable, with a maximum speed of 0.2 seconds and a minimum speed of 2 seconds. Within this range, the speed could be adjusted as needed to meet specific usage requirements.

4.4 Degrees of Freedom

Each finger of the exoskeleton could be controlled independently or in any combination, and the thumb module achieved arbitrary movement in two degrees of freedom, shown in figure 14. The thumb could perform adduction/abduction and flexion/extension movements as well as composite movements, fulfilling the additional bonus objectives set forth in the design criteria.

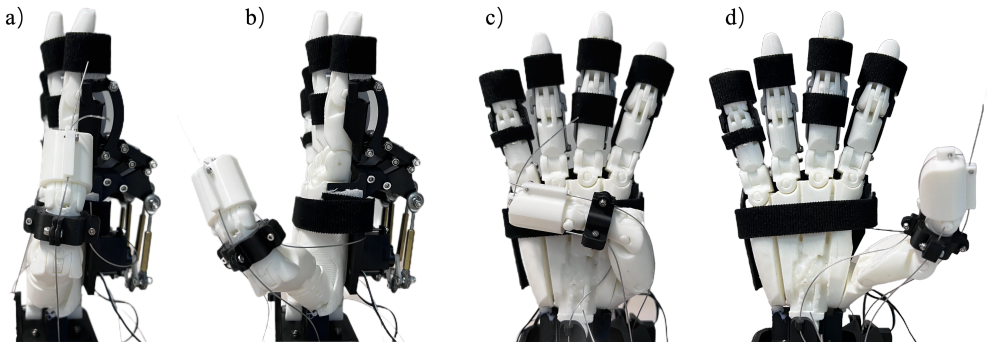

Figure 14: Schematic of the thumb module with two DOF.

4.5 Comfort and Portability

The exoskeleton was designed to be comfortable and portable, with a total device weight of only 400 grams, making it easy to wear and minimizing the burden on the user. The electronic control and power supply components were directly fixed to the exoskeleton, eliminating the need for an additional backpack and enhancing user convenience .

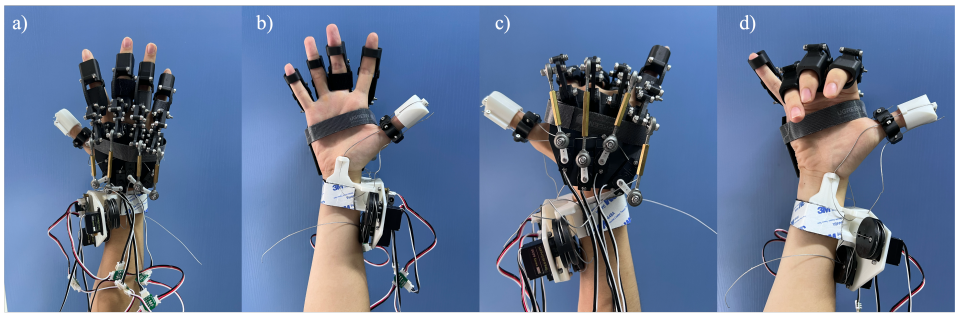

Figure 15: Schematic of the wearable test for two actions (number 5 and 6).

We conducted a wearability test of the exoskeleton on a person's hand to evaluate its ease of comfort and functionality. During the test, the exoskeleton demonstrated excellent wearability, with the Velcro straps allowing for quick and secure fastening, making the device easy to put on and take off. The adjustable design provided a comfortable fit, accommodating different wrist and finger sizes

without causing discomfort. Additionally, the exoskeleton effectively enabled finger bending movements, demonstrating its practical application for assisting with hand rehabilitation and daily activities, shown in figure 15.

These results indicate that the hand-assist exoskeleton meets or exceeds the initial design specifications and functional requirements, offering a reliable and effective solution for hand rehabilitation and assistance.

4.6 Finite Element Analysis (FEA)

The FEA of the finger linkage structure reveals significant stress concentrations, particularly in the highlighted areas. These results suggest that the current material and design are susceptible to mechanical failure under operational loads. To enhance structural integrity, future iterations should consider using fiberglass-reinforced polymer (FRP) for its superior strength-to-weight ratio, optimizing the linkage geometry to distribute stress more evenly, and employing advanced manufacturing techniques to improve durability and performance. After changing the material, the stress is reduced and more evenly distributed.

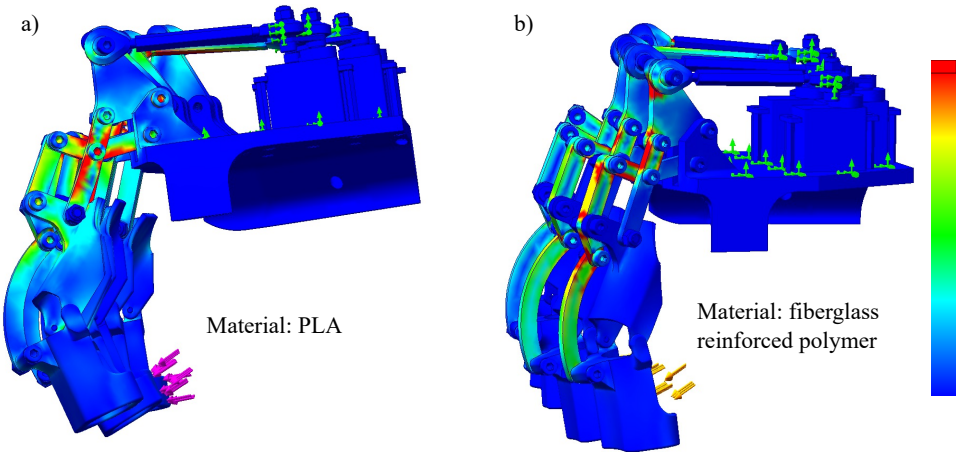

Figure 16: Schematic of inite Element Analysis (FEA).

5 Cost Estimation

The total project cost was carefully calculated by excluding certain unused materials and considering cost-saving measures. The final budget required to realize this project, without accounting for batteries, 3D printing, welding materials and tools, mechanical consumables and tools, nylon rope, fishing line, and testing iterations, is 397 CNY.

Therefore, this budget ensures the project's financial feasibility while maintaining the required quality and functionality for effective hand rehabilitation. Future expenses will focus on refining the device through iterations and integrating advanced technologies to enhance its capabilities.

6 Conclusion

This study introduces a novel exoskeleton for hand rehabilitation that combines rigid-tendon mechanisms to enhance the efficacy and consistency of therapeutic outcomes. The exoskeleton, designed to address the needs of stroke and traumatic brain injury survivors, integrates advanced mechanical design with high-precision control technologies to facilitate effective rehabilitation training.

Our device, which includes a rigid finger module utilizing a spatial four-bar linkage mechanism and a tendon-driven thumb module, demonstrates significant advancements in safety, functionality, and user comfort. The integration of bus servo technology with closed-loop control capabilities ensures precise and reliable movements, critical for effective rehabilitation. Additionally, the use of additive manufacturing with PLA material provided a balance of strength, precision, and cost-effectiveness, while the user-friendly UI interface enhanced the device's accessibility and usability.

The exoskeleton's performance was validated through extensive testing, demonstrating its ability to safely replicate natural hand movements and generate a maximum gripping force of 23.64 N, surpassing the initial design requirements by 18.2%. The device offers customizable movement speeds ranging from 0.2 to 2 seconds, with independent or combined control of each finger's movements. The thumb module achieves two degrees of freedom, enabling complex thumb movements essential for comprehensive hand rehabilitation.

With a total weight of 400 grams, the exoskeleton is lightweight and portable, making it a practical solution for both medical rehabilitation and assistive applications. The device's compact and integrated design enhances user comfort and convenience, supporting its potential for widespread use.

Future work will focus on integrating advanced technologies such as the Internet of Things (IoT) and cloud computing to enhance the system's intelligence and personalization. Conducting clinical trials and refining the design based on user feedback will be crucial steps in further developing and validating the exoskeleton's effectiveness. By leveraging these advancements, we aim to provide a more intelligent, personalized, and effective rehabilitation solution that can significantly improve the quality of life for individuals recovering from hand impairments.

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